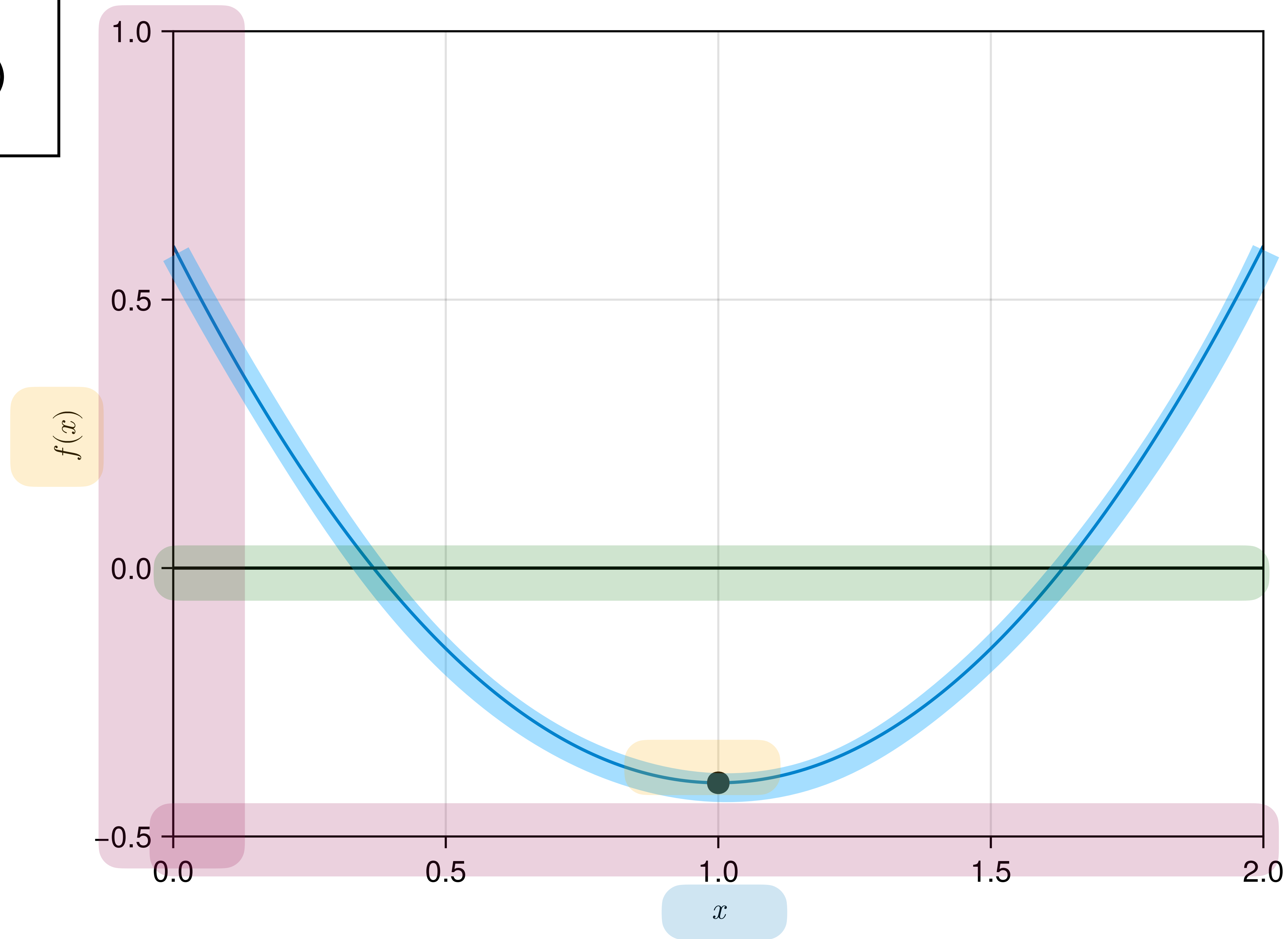
```julia
using CairoMakie
f(x)=x^2-2x+0.6;
xplot=0:0.01:2


fig = Figure()
ax = Axis(fig[1, 1],
     xlabel = L"x",
     ylabel = L”f(x)",
     limits=(0,2,-0.5,1.0))
lines!(ax,xplot,f.(xplot)) #main code to plot f(x)
hlines!(ax,[0],color=:black)
scatter!(ax,1,f.(1);color=:black,markersize=15)
save("../figures/01p03PlottingFunctionsInJulia01.svg", fig)
display(fig)
```

$$f(x) = x^2 - 2x + 0.6$$

$$\begin{Bmatrix} 0.00 \\ 0.01 \\ 0.02 \\ 0.03 \\ \vdots \\ 2.00 \end{Bmatrix} \qquad \begin{Bmatrix} 0.00^2 - 2 * 0.00 + 0.6 \\ 0.01^2 - 2 * 0.01 + 0.6 \\ 0.02^2 - 2 * 0.02 + 0.6 \\ 0.03^2 - 2 * 0.03 + 0.6 \\ \vdots \\ 2.00^2 - 2 * 2.00 + 0.6 \end{Bmatrix}$$



```julia
himmelblau(x)=(x[1]^2+x[2]-11)^2+(x[1]+x[2]^2-7)^2

x1range=-6:0.02:6;
x2range=-6:0.02:6;
funcplot = [himmelblau([x1,x2]) for x1 in x1range, x2 in x2range]

fig = Figure()
ax = Axis(fig[1, 1],
    xlabel = L"x_1",
    ylabel = L"x_2",
    xlabelsize = 20,
    ylabelsize = 20,
    title = "Himelblau function",
    limits=(-6,6.0,-6.0,6.0))
levels = 10.0.^range(0, 3.5; length=100)
contour!(ax,x1range,x2range,funcplot; levels,color=:black)
contourf!(ax,x1range,x2range,funcplot; levels,colormap=:bwr)
save("../figures/01p03PlottingFunctionsInJulia02.svg", fig) #save figure as svg file
display(fig)
```

$$h(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

$$h(x, y) = (x^2 + y - 11)^2 + (x + y^2 - 7)^2$$

$$-6 \le x_1, x \le 6$$

$$-6 \le x_2, y \le 6$$

$$\longrightarrow \begin{Bmatrix} -6.00 \\ -5.98 \\ \vdots \\ 5.98 \\ 6.00 \end{Bmatrix}$$

$$\begin{bmatrix} h(x_1, y_1) & h(x_2, y_1) & \dots & h(x_N, y_1) \\ h(x_1, y_2) & h(x_2, y_2) & \dots & h(x_N, y_2) \\ \vdots & & & \\ h(x_1, y_N) & h(x_2, y_N) & \dots & h(x_N, y_N) \end{bmatrix}$$